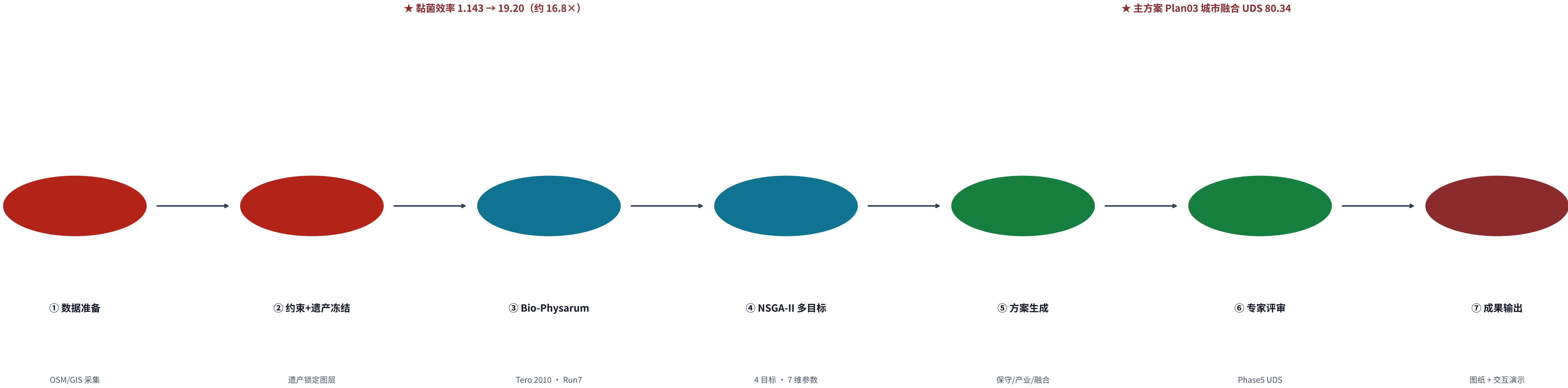
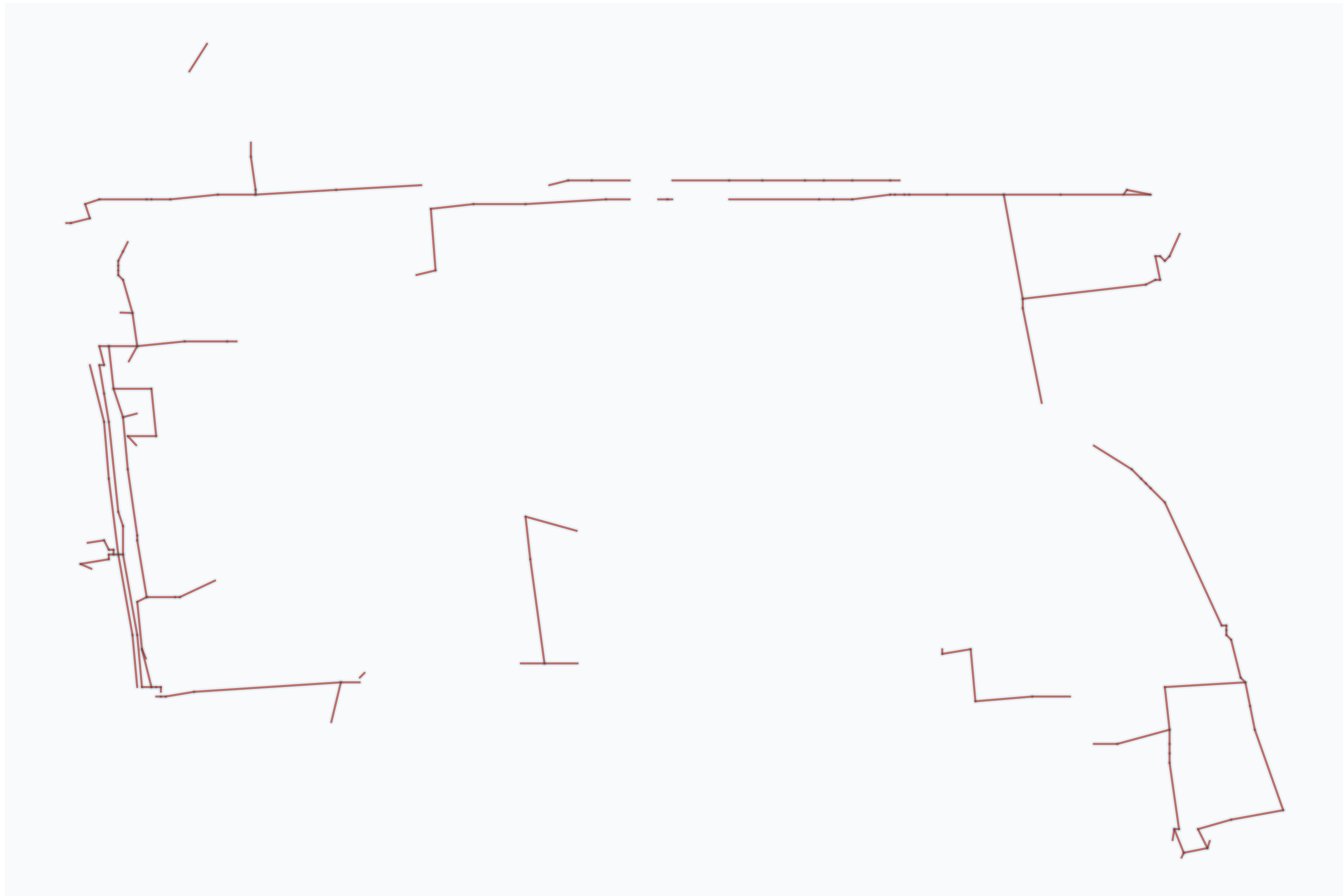


Method & Results: Bio-Physarum Adaptive Network + NSGA-II

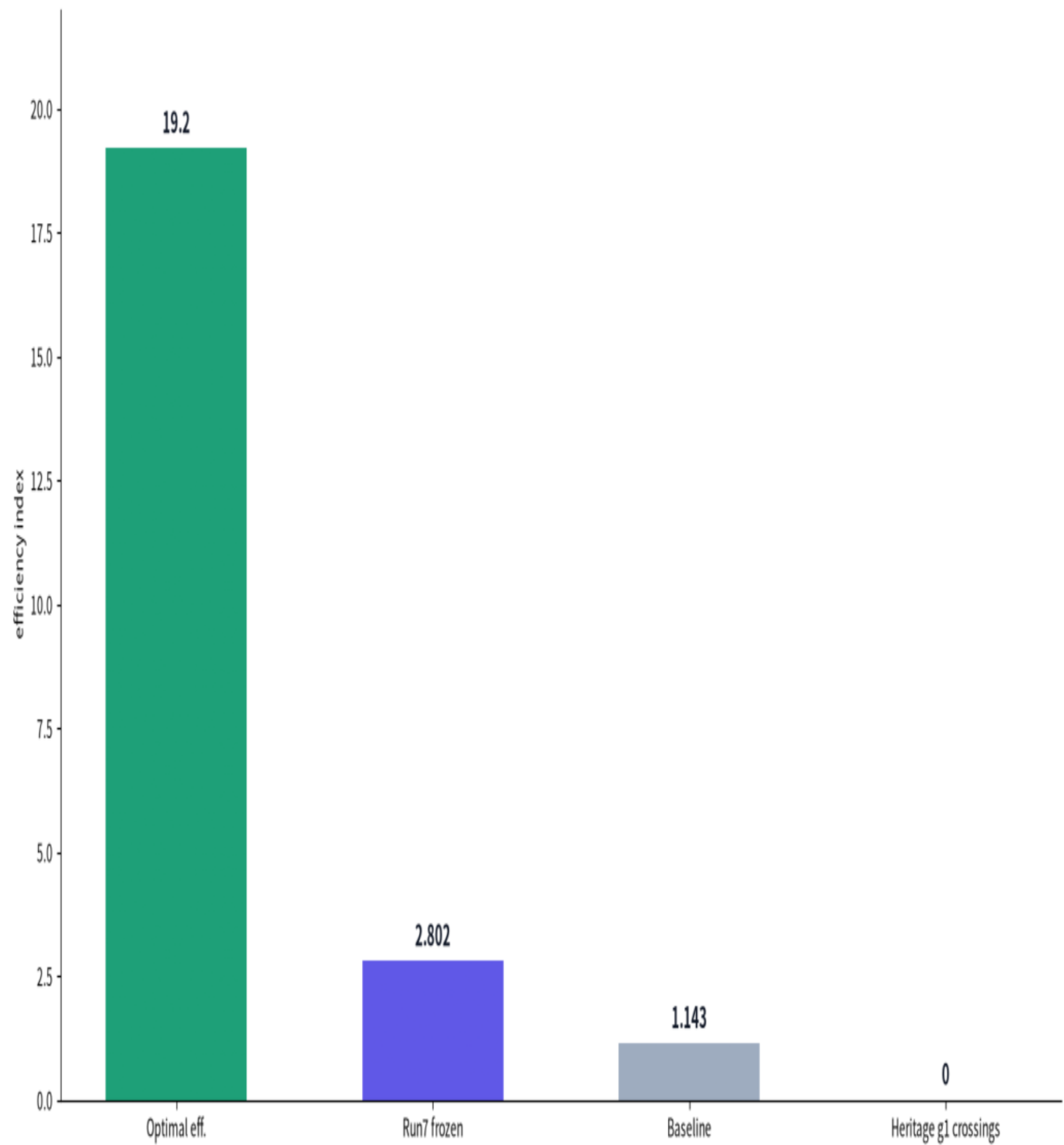
Technical route (Phase 0 → deliverables)

Real Physarum 167-edge skeleton (method validation)



Core metrics & method validation

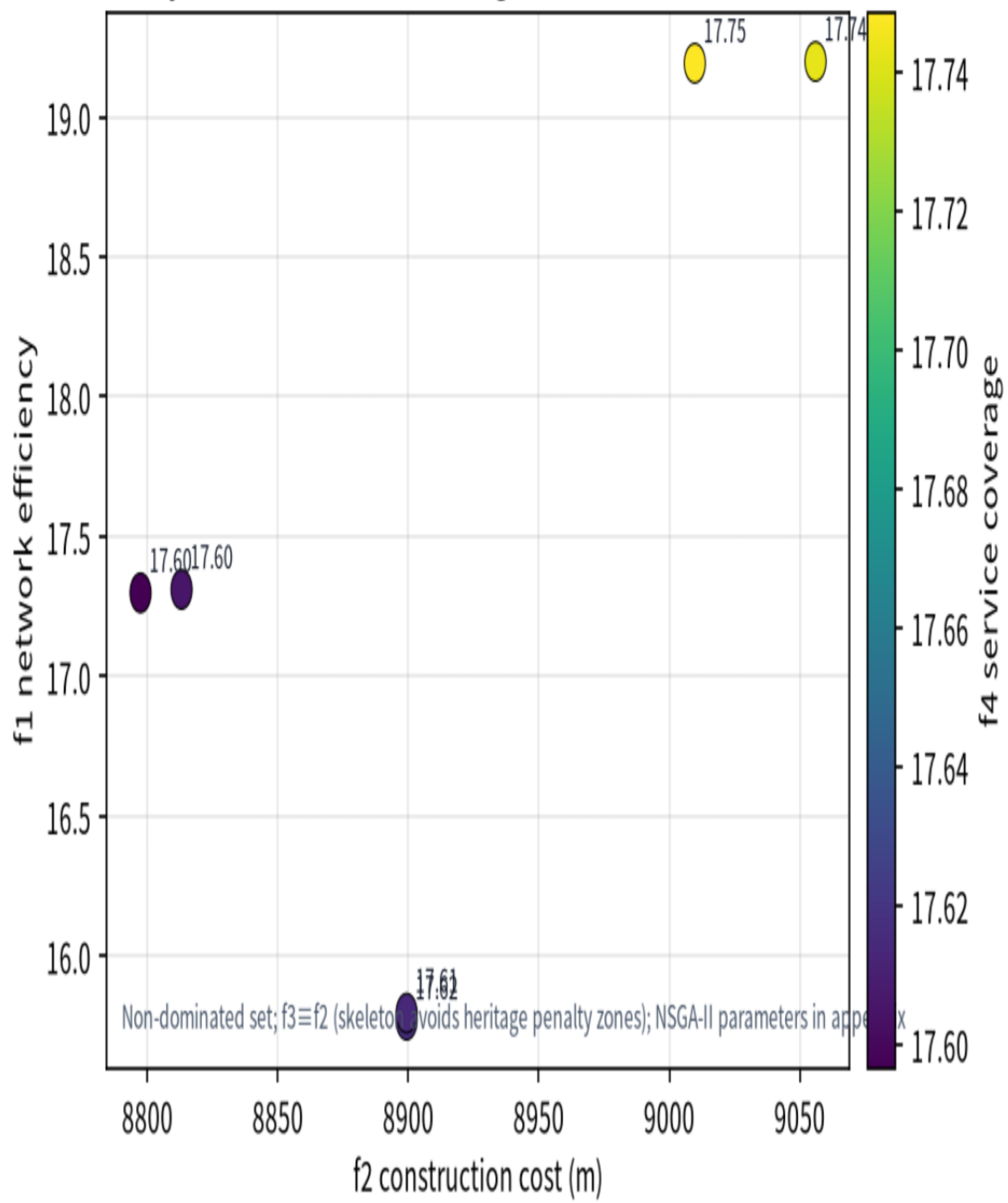
Metrics and method validation



Plan03 urban-integration UDS 80.34 recommended; heritage objective UDS=0 degenerates to cost — "0 heritage crossings" is g1 hard-constraint behavior only, not an independent site heritage conclusion
Method validation (real Physarum run): 167-edge network / efficiency 19.20 / heritage objective UDS=0 (degenerates to cost; no independent site heritage conclusion)

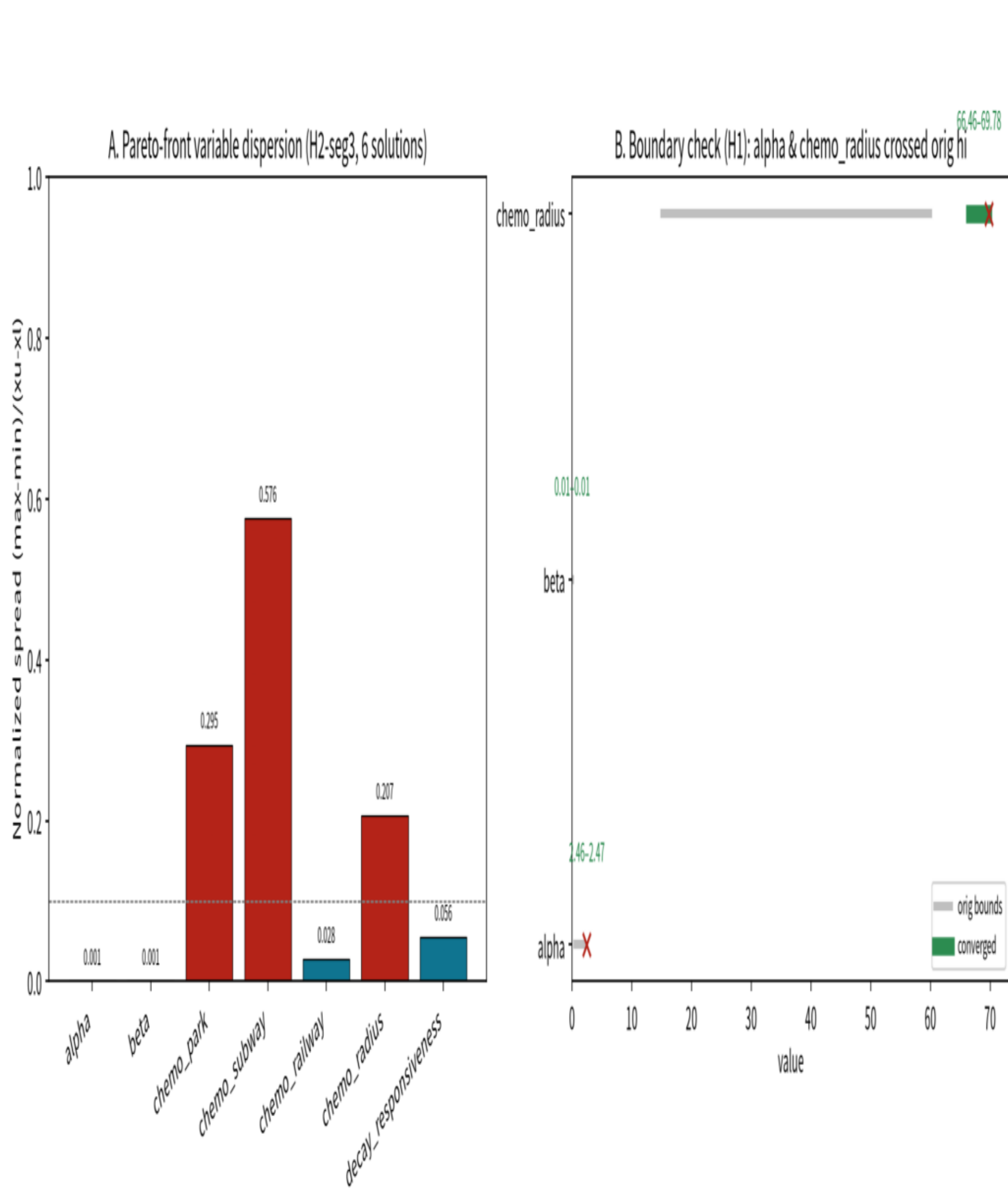
Four-objective Pareto front (6 sol.)

Four-objective Pareto front (H2-seg3, 6 non-dominated solutions)



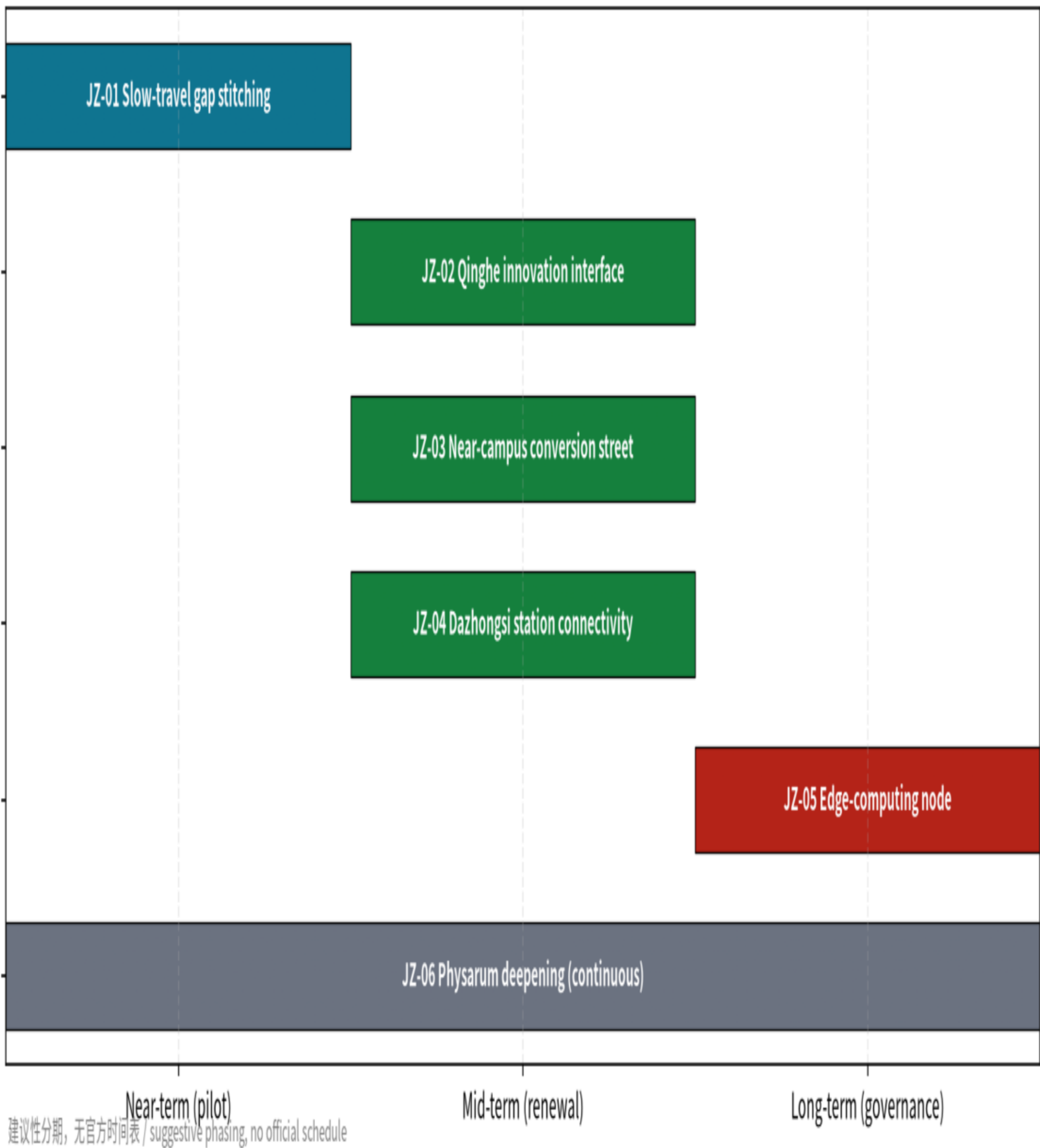
Parameter sensitivity (suggestive)

Parameter sensitivity – Bio-Physarum + NSGA-II (real runs, method validation)



Suggestive phasing gantt (6 projects × 3 phases)

Suggestive phasing gantt (6 projects x 3 phases)



建议性分期, 无官方时间表 / suggestive phasing, no official schedule

19.20

Optimal eff.

1.143

Baseline

2.802

Run7 frozen

167

Skeleton edges

8813 m

Skeleton length

80.34

Plan03 UDS

69.1%

Permeable

25.2%

Green pen.